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Machine Learning Techniques for Predictive Analytics of Academic Outcomes and Behavior of Students

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Abstract— **This study looks at how machine learning (ML) can be used to look at and guess how well secondary school pupils in two Portuguese schools will do in school. The study uses a dataset with 33 variables that include a lot of demographic, social, and academic elements, such as age, gender, study time, family history, and the education levels of the parents. To make it easier to sort the data, the objective variable students' final grade (G3) was changed into a binary outcome: students who scored 10 or higher were put in the "pass" (1) group, and those who scored less than 10 were put in the "fail" (0) group. We used three machine learning models which are: Random Forest, Logistic Regression, and Gradient Boosting to see how well they could predict. Random Forest had the best accuracy of the three models, at 92.4%, followed by Logistic Regression at 89.9%, and Gradient Boosting at 86.08%. The study shows that past academic performance and study habits are important indicators of how well a student will do. The results show that machine learning could be very useful in schools, which is helpful information for teachers, school administrators, and politicians. These ideas can help build focused interventions, personalized learning plans, and early warning systems that will help students do better and lower the number of students who fail in secondary school.**

Keywords—machine learning, student performance, classification, logistic regression, student behavioral pattern, artificial intelligence

I. Introduction

Education plays a significant role in the development of both individuals and societies, making it an essential component for every individual [1]. The educational systems in Portugal continue to encounter challenges, with high failure rates in fundamental topics such as mathematics and Portuguese, despite the fact that there have been significant advancements in these teaching methods. The early school departure rate for students between the ages of 18 and 24 in Portugal was forty percent in 2006, which was much higher than the average of fifteen percent for the European Union [2]. These trends underscore the importance of addressing factors influencing academic performance to implement effective interventions [3].

Through the course of this research, the impact of a wide range of demographic and socioeconomic factors on academic achievement is brought to light. The results of a meta-analysis conducted by Fan and Chen indicate that there is a favorable correlation between parental participation and the achievement of students [4]. Similarly, Wang and Eccles demonstrated the critical role of social support in fostering academic engagement and performance [5]. Extracurricular activities, such as participation in student councils or sports also contribute to improved academic outcomes [6]. However, comprehensive studies examining these variables in the Portuguese educational context are limited. The integration of Business Intelligence (BI) and Data Mining (DM) in education has been widely studied for its ability to transform raw data into actionable knowledge. The classification techniques are widely used in educational data mining, enable accurate predictions of student performance based on structured data [7]. Among these, the Random Forest algorithm is recognized for its robustness and accuracy in handling complex datasets by aggregating multiple decision trees [8]. This research addresses two major challenges are to clarify the complex relationships between various factors affecting academic performance and aims to provide educators and policymakers with data-driven insights for improving student outcomes. Understanding these relationships is crucial for making informed long-term decisions regarding educational strategies and resource allocation. These insights are essential for improving academic achievement and addressing systemic educational issues in Portugal. The primary objectives of this study are: (i) Analyze correlations between demographic, social, and academic features with student performance, (ii) Identify Influential Factors: Determine the key predictors of high or low academic achievement and (iii) Utilize advanced data mining techniques, including Random Forests and Support Vector Machines to forecast final grades (G3) based on prior grades (G1, G2) and other variables.

II. Literature Review

A. Text Analysis

Text analysis involves splitting and analyzing text to derive meaning in fields like education, business, or healthcare. Word cloud and topic modelling are complementary approaches to exploring textual information. Word clouds, created using Python's word cloud library, find commonly occurring terms. Preprocessing steps like converting text to lowercase, removing punctuation,

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numbers, and stopping words ensure focus on meaningful terms. Word clouds provide a visual overview of key themes [9, 10]. Word clouds give a visual representation of top terms but are less clear on frequency distribution compared to bar plots, which quantitatively represent term frequencies[11]. Bar graphs visually compare data across categories using rectangular bars proportional to their values. Latent Dirichlet Allocation (LDA) and topic modelling uncover deeper patterns in textual data. Topic modelling groups document collections into topics without pre-existing knowledge. LDA, using a "Bag of Words" approach, analyses word frequencies to reveal themes. It outputs topics as word distributions and probabilities for documents. These techniques combine numerical and qualitative insights. Topic modelling uncovers patterns, while word clouds offer quick focus points. Together, they bridge quantitative and qualitative data, extending analysis depth and providing actionable information [12, 13].

B. Classification

Classification is a supervised learning technique, in machine learning for predictive modelling for labelling an example based on a class label. It is a function that maps input variables to output variables: target labels or categories. Structured and unstructured data can be classified as well as spam in email services where emails are categorized as "spam" or "not spam." This work provides an overview of the typical classification problems. Binary classification: This involves two class labels which can be 'true/false', 'yes/no'. The abnormal state is represented by another class while the normal state is represented by another. In a medical test, for example, “cancer not detected” could be normal and “cancer detected” abnormal. Multiclass classification: Soft biometric recognition that engages additional than two class labels but is not scattered into normal and abnormal dynamics. For instance, data in the NSL-KDD set of network connections may be categorized into; DoS, U2R, R2L, and Probing attack. Multi-label classification: This includes the examples where an instance is related to more than one class; the classes are not distinct. In fact, one sample may relate to many classes at the same time, for instance, Google News categorizes articles and labels them under city name, technology, and latest news at the same time [14].

1) Random Forest

Random Forest (RF) is a machine learning technique consisting of a number of decision trees combined for reducing correlation in feature data. The Random Forest computational cost O(n) (n is number of samples) makes it easy to use it with large datasets. Furthermore, RF can be easily parallelized because processing time is significantly reduced. As random Forest relieves the correlation between decision trees, the random selection of samples and features is used. To begin with, an amount equivalent in magnitude to the entire dataset is randomly selected from the training dataset. Additionally, each decision tree is built using a subset of features that are randomly chosen. These two forms of randomization reduce the correlation between individual decision trees, reducing the risk of overfitting, and improving the model’s accuracy. A model that uses the random forest algorithm was build and applied on the learning data of 395 students in the Chongqing University of Posts and Telecommunications Python Programming course and was used to predict and analyze student performance. The results, as shown in Figure. 1 below, from evaluation show the model can accurately predict student performance ($R^2 = 0.55$, RMSE = 4.68 points, MAE = 3.61 points).

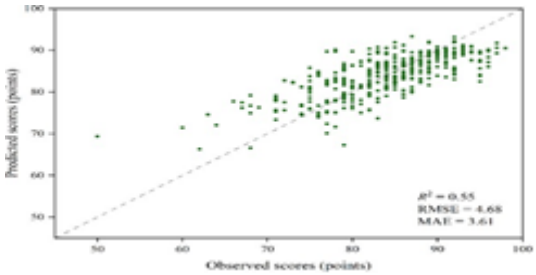

Fig. 1. Comparison of Predicted and observed scores

2) Logistics Regression

Logistic regression is widely used to analyze binary response data which mainly involves success/failure values which are often coded 1/0. This is different from standard linear regression which costs normality and homoscedasticity, logistic regression originates from Bernoulli distribution and therefore does not have biases when estimating binaries. It lies within the family of exponential models for the ease of estimation of the parameters needed in its models. The estimation is based on the log-likelihood function so one can sum over observations instead of multiplying. Apparently, the generalized linear model (GLM) algorithm for maximum likelihood estimation can apply logistic regression and other models like the probit and poisson models. The academic performance of students has been predicted to use OLR by a study as shown in Figure. 2 below and the ultimate grade is the dependant variable. Pass, Third- Class, Second-Class Lower, Second Class Upper and First Class were the categories of grades. The variables analysed were mode of entry, age at entry, sex and department.

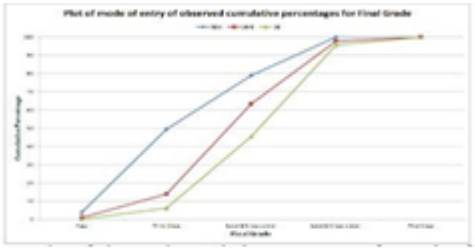

Fig. 2. Plot of Observed Cumulative Percentages for mode of entry

The findings show the mode of entry matter regarding the chances of hitting higher grades. Concretely, students admitted through Remedial (REM), Unified Tertiary Matriculation Examination (UME), and Direct Entry (DE) were most likely to graduate with First Class while students admitted through Remedial (REM) and Unified Tertiary Matriculation Examination (UME) are less likely. DE students are matured most of the time, having been with their previous school for at least 2 academic sessions and most of them went to Polytechnics.

3) Gradient Boosting

Gradient boosting is a popular ensemble method in which many weak models are fitted sequentially to do so that the final model is stronger than each of the weak ones. It gives

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rise to an ensemble of weak, base learner models such as decision trees, and the ensemble is iteratively built. At each stage, the new model is trained to be maximally correlated with the negative gradient of the loss function, thereby reducing error. Gradient boosting machines in essence are first order approximators for capturing complex non-linear conditions and are highly customizable to fit many different needs. Nevertheless, their main limitation is high memory consumption, as storing a predictive model depends on the number of boosting cycles and often needs tens of thousands of iterations. Sparse base–learners or ensemble simplification strategies can help to some degree mitigate this. The other is assessment speed, as evaluating all base-learners in a large ensemble is slow. Though parallelization boosts prediction speed, the gradient boosting is sequential in training as opposed to the random forest where parallelization is faster overall. While these are challenging, flexibility enables professionals to define models for particular tasks and customize them to multiple loss functions and data specific base learners.

4) Support Vector Machine

Support Vector Machine abbreviated as SVM was invented in 1992 by Boser, Guyon and Vapnik in COLT-92 conference and was used in the binary and multiclass predictors and regressors. They belong to the family of generalized linear classifiers and the hypostasis space is the optimised and trained linear functions of possibly high dimension feature space discovered with optimization and statistical learning theory. Classification with SVM is achieved by the identification of a hyper-line in a higher dimensional space. Even more specifically, it means that the term hyper-plane is intended here by the decision planes. A decision plane is that which divides a group of data of a particular type from the other group of data of the other type. Any suitable nonlinear mapping for higher dimensional space always performs classification by demonstrating a hyper-plane. SVM, thus, considers those vector points, known as the support vectors or those points that determine the decision boundary and offer the large marginal with regards to the mentioned two classes. The advantages of Support Vector Machines (SVMs) include high generalization capability, which arrests over- fitting by optimizing the margin of the different boundaries or decision planes. It is able to handle small data sets, and in storing it, it uses support vectors [15, 16].

5) K-Nearest Neighbors (k-NN)

K-Nearest Neighbors (k-NN) is one of the easy-to-implement classifier methods applied for machine learning. It categorizes data points using the nearest class of a given data set or elements of the same data set [17]. The approach operates in two main stages: in the case of the k nearest neighbors, first assigning the k closest samples to a query example, and second, defining the class of the query based on its neighbors utilizing the majority rule or the weighted majority rule [17]. The k-Nearest Neighbor (k-NN) algorithm is easy to implement, independent of the scale of input, and flexible for the choice of norm, making it suitable for classification model. Unlike some other clustering techniques, it can accommodate data that are not feature vectors, but it requires condition that an adequate similarity metric is determined and so it can successfully be used in a broad range of problems including text categorization, image recognition, and time- series analysis. Forecasts are made considering the proximity to the data points of the same group, which simplifies interpretation.

6) Naïve Bayes

The Naive Bayes classifier is the machine learning model for classification problems and makes the classification depending on the probability distribution of a dependent variable and independent variables, believing that the inputs are independent conditioned by the output [18]. If given an input, it is rather able to predict how often a given class will be obtained using the Bayes' theorem. The Naive Bayes classifier is regarded as simple, fast and accurate, therefore frequently utilized in the classification processes especially in the text categorization and spam filtering [19]. Its computational tractability is due to the CI assumption, which makes estimation less complex and achievable in high-dimensional feature spaces. It has probabilistic output that presents the degree of uncertainty as a confidence score beneficial for uncertainty management in decisions. The model's simplicity enables interpretation and application of the results particularly when working with small datasets. Laplace correction and Dirichlet estimation and other methods try to overcome the problem of zero probabilities and is more accurate in case of meager data. These features make Naive Bayes to be deemed a fast and accurate classifier specifically when applied on textual datasets.

III. METHODOLOGY

This research uses a student performance dataset by Paulo Cortez [20]. The dataset is derived from secondary education students in two Portuguese schools through school reports and questionnaires. It includes 33 attributes encompassing demographic, social, and academic features, such as student age, gender, study time, and parental education level [21]. Before using the dataset for our research, we will process it manually with tools to ensure that the data aligns with our objective and is ready to be used for our experiment. The dataset will go through cleaning and filtering, feature encoding and feature scaling. The dataset will be reviewed to ensure data integrity such that there is no missing values. If any attributes were found to have missing entries, it will be analyzed, and appropriate actions will be taken. However, no significant cleaning was needed as the dataset was found to be well-structured without any missing value.
Next feature coding would be done. After the dataset was cleaned and checked, the categorial features such as "gender" and "school", were converted into binary numerical representations using one-hot encoding. This is to ensure that algorithms that require numerical input will be able to understand the data. As for ordinal features, such as "parental education level", it will be encoded to preserve their inherent order.

Next feature scaling would be done. As for numerical features, such as "study time" and "absences", it will be standardized using StandardScaler. This is to help with models that are sensitive to feature scaling such as logistic regression and gradient boosting. The target variable (G3') represents the final year grade. To align with the binary classification task, 'G3' will be transformed into a binary

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variable, where students with grades ⩾ 10 will be labeled as "pass" (1), and those with grades < 10 will be labeled as "fail" (0). This transformation allows for the classification of students into two distinct categories of academic achievement.

Before running the experiment, the dataset will be divided into two subsets, 80 percent for training and 20 percent for testing using train_test_split with a random state for reproducibility. This ensures that the model can learn to account for unforeseen data, reducing the risk of overfitting. The following algorithms will be used for binary classification tasks. Each of the algorithms chosen will be trained using the same postprocessing dataset and be evaluated equally.

A. Logistic Regression:

A type of statistical model that is used for classification and predictive analysis. The model uses a sigmoid function to map all arguments between the range of 0 and 1, classifying binary outcomes. Then a decision boundary threshold will be set to classify the input into discrete classes based on the predicted probability [22]. This algorithm is chosen for its binary classification ability, scalability, and interpretability. It is also relevant as it can help classify and predict based on one or more predictor variables.

B. Random Forest:

This classifier uses an ensemble learning method as it concludes its decision based on multiple decision trees. Thus, improve predictive accuracy and helps with overfitting [23]. This classifier is chosen for its classification, high dimensionality, and precision. This classifier is relevant as it can classify better with high-dimensional data.

C. Gradient Boosting

This classifier also uses ensemble learning method to build models sequentially, where each new model corrects the previous one [24]. Gradient Boosting is chosen for its classification, predictive accuracy, and ability to handle complex datasets. It is relevant as it can constantly improve itself to provide a more accurate result.

To evaluate model performance, a combination of quantitative metrics was employed:

- Accuracy: Measures the proportion of correct predictions as to the total predictions·
- Confusion Matrix: Provides a detailed breakdown of the model's performance. It shows the counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This insight helps us understand the model's error rate, strengths and weaknesses.
- Classification Report: Provides a comprehensive report of precision, recall, F1-score, and support for each class. These metrics collectively evaluate a model's overall performance and its ability to correctly identify each class.

IV. Result & Discussion

Random Forest achieved the highest overall accuracy among the three models, excelling in both precision and recall. Its ability to utilize multiple decision trees likely contributed to its superior performance, as it can better capture the relationships between features. The high importance of features like G2 and G1 demonstrates the model's reliance on academic performance in previous periods, aligning with the target variable G3. This result is shown in Figure 3.

TABLE I. Performance Evaluation Metrics of ML Techniques

ML Techniques	Accuracy	Precision	Recall	F1-score
Random Forest	92%	98%	90%	94%
Logistic Regression	90%	94%	90%	92%
Gradient Boosting	86%	90%	88%	89%

```
Random Forest
Accuracy: 0.9240506329113924
Confusion Matrix:
 [[26  1]
 [ 5 47]]
Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.96      0.90        27
           1       0.98      0.90      0.94        52

    accuracy                           0.92        79
   macro avg       0.91      0.93      0.92        79
weighted avg       0.93      0.92      0.93        79
```

Fig. 3. Results of Random Forest Model

Logistic Regression shown in Figure 4, provided robust results, particularly in terms of precision for the "Pass" class. Its slightly lower accuracy compared to Random Forest could be attributed to its linear decision boundary, which may not capture complex relationships between features as effectively as ensemble methods.

```
Logistic Regresion
Accuracy: 0.8987341772151899
Confusion Matrix:
 [[24  3]
 [ 5 47]]
Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.89      0.86        27
           1       0.94      0.90      0.92        52

    accuracy                           0.90        79
   macro avg       0.88      0.90      0.89        79
weighted avg       0.90      0.90      0.90        79
```

Fig. 4. Results of Logistic Regression Model

Gradient Boosting, as seen in Figure 5, while achieving the lowest accuracy among the three models, demonstrated consistent performance across metrics. Its ability to iteratively correct errors from previous trees allows it to capture subtle patterns in the data. However, further tuning of hyperparameters, such as learning rate and number of estimators, could potentially enhance its performance.

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```
Gradient Boosting
Accuracy: 0.8607594936708861
Confusion Matrix:
 [[22  5]
 [ 6 46]]
Classification Report:
              precision    recall  f1-score   support

           0       0.79      0.81      0.80        27
           1       0.90      0.88      0.89        52

    accuracy                           0.86        79
   macro avg       0.84      0.85      0.85        79
weighted avg       0.86      0.86      0.86        79
```

Fig. 5. Results of Gradient Boosting Model

A comparative analysis of all the models shows that Random Forest demonstrated the best overall performance, achieving the highest accuracy (92.4%) and a strong balance between precision and recall across both classes. Logistic Regression also performed well, with an accuracy of 89.9% and high precision for the "Pass" class. Gradient Boosting, while slightly less accurate (86.1%), provided robust performance metrics and may benefit from hyperparameter tuning for further improvements. These findings highlight the importance of selecting appropriate algorithms and the potential for ensemble methods like Random Forest in binary classification tasks as seen in Figure 6.

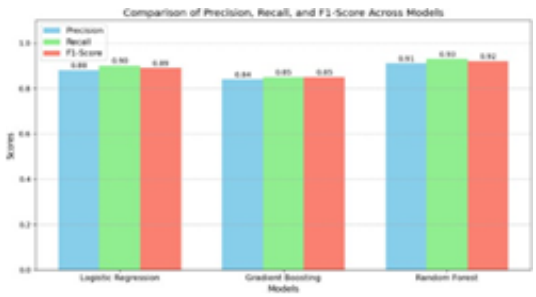

Fig. 6. Comparative Analysis using Evaluation Metrics for RF, GB and LR

The purpose of this study was to investigate and forecast the level of academic accomplishment among students enrolled in secondary education at two Portuguese institutions. To do this, the researchers utilized a variety of machine learning algorithms for categorization tasks. The datasets that were utilized included a wide variety of demographic, socioeconomic, and academic characteristics, with the final grade (G3) serving as the variable that was the focus of attention. The G3 was converted into a pass/fail outcome based on a threshold grade of 10, which was done so that binary categorization could be used more easily.

V. Conclusion

From the study, Random Forest achieved the maximum accuracy of 92.4%, while Logistic Regression achieved the accuracy of 89.9%, demonstrating their usefulness in classifying student performance. The findings of the research suggested that Random Forest achieved the highest accuracy. It was discovered through the confusion matrices that both models had comparable performance metrics, with good precision and recall for predicting students who were successful in passing the test. It is important to note that Gradient Boosting achieved a lower accuracy of 86.1%, which highlights the fact that it is sensitive to the configurations of the hyperparameters. The investigation has demonstrated that Random Forest is a reliable method since it successfully controlled the intricate linkages that existed within the data while simultaneously preserving a low rate of misclassification error. Furthermore, the interpretability of the Logistic Regression, which made it a great tool for analyzing the influence of many factors on student outcomes without sacrificing its predictive strength, was another advantage of the method. Despite the fact that Gradient Boosting displayed a little diminished performance, the iterative nature of the algorithm suggests that additional tuning could potentially result in improved outcomes. In general, the findings of this study reveal that the application of machine learning techniques has the potential to considerably improve our comprehension of the dynamics of student performance in educational environments. Taking into account other relevant factors like parental education and study habits, the findings highlight the significance of adding previous academic performance (G1 and G2) as key predictors. Additionally, the findings highlight the necessity of incorporating these aspects. Educators and policymakers can build tailored interventions to improve student outcomes and maximize resource allocation within schools by utilizing these findings and putting them to use. In addition, the work that will be done in the future can concentrate on further refining the parameters of the model and investigating additional features in order to improve prediction capabilities even more.

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